

Equipo utilizado para medir la Demanda Bioquímica de Oxígeno (DBO) en muestras de agua. Permite evaluar la cantidad de oxígeno consumido por microorganismos durante la descomposición de materia orgánica.

**DBO DIGITAL
SENSOR ROTATIVO**



Respirometric Systems

Operating Manual



RESPIROMETRIC Sensor System

Equipment for respirometric measurement.

Operating Manual

Thank you for choosing the VELP RESPIROMETRIC Sensor System.

SAFETY WARNINGS



Before using the instrument, it is recommended to carefully read the operating manual supplied with the equipment. Improper use of the device that is not in accordance with the instructions in the manual can compromise its safety.



The instrument falls into the category of electrical and electronic equipment and therefore in the event of definitive decommissioning it must be subjected to separate collection and cannot be disposed of as urban waste, in accordance with the provisions of Directive 2002/96 /EC. For more information on recycling this product, contact your local municipal office.

This instrument is to be used for laboratory applications only and for internal use only.

SAFETY

To move the device, especially when the bottles with the samples are inserted, use the special recessed handles on the sides of the instrument. Avoid moving or carrying the instrument by holding it elsewhere. Magnetic fields can exert their effects on magnetic or metal pieces (magnetic media, floppy disks, pacemakers, magnetic cards). Keep these pieces away from the magnetic stirrer and magnetic stirrers.

CLEANING

Cleaning the instrument must be carried out, after disconnecting the mains power supply, with a cloth moistened with non-flammable and non-aggressive detergents.

MEANS OF PERSONAL PROTECTION

The means of personal protection must be compatible with the risks deriving from the reagents necessary for the analytical procedures and from the glass bottles.

MAINTENANCE

For routine maintenance of the appliance, see the Maintenance chapter in this instruction manual.

In accordance with the product warranty law, all repairs to our tools must be carried out at our facility, unless otherwise agreed upon with local distributors.

The instrument was designed and constructed in accordance with the following standards:

Safety requirements for electrical equipment for
measurement and control and for laboratory use
Electrical equipment for laboratory use
General Requirements - Canadian Electrical Code

IEC/EN61010-1
UL 3101-1
CAN/CSA C22.2 No 61010-1

The manufacturer, in the commitment to constantly improve the quality of its products, reserves the right to vary specifications without notice.

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1. Introduction

The design, monitoring and control of biological treatment is today based on the knowledge of kinetics and stoichiometry of the bioprocesses involved in biological treatment. To this end, respirometric techniques have been developed to monitor aerobic and anaerobic processes.

In the BOD test, the microorganisms present or specially inoculated in a water sample containing a biodegradable organic substance consume oxygen for their metabolic activity and produce a corresponding volume of carbon dioxide. If one operates in a closed system and the carbon dioxide produced is absorbed by a highly alkaline substance (soda lime, caustic potash), there will be a progressive decrease in the gas pressure, which can be measured with suitable manometers.

As for the BMP (Biochemical Methane Potential) analysis, it is used to determine the methane potential and the biodegradability of waste water and biomass waste using anaerobic bacterial culture. During the test, the production of biogas and / or methane is monitored over time, which is proportional to the quantity of organic substance gradually degraded.

VELP Scientifica has specially developed, in accordance with international safety regulations, the innovative RESPIROMETRIC Sensor electronic system that can be used with stirring systems that allow the simultaneous housing of 6 bottles.

The RESPIROMETRIC Sensor, which screws directly onto each bottle, is equipped with advanced electronics. The pressure transducer controlled by a microprocessor allows the storage of BOD and pressure values. The data is transmitted wirelessly from the sensors to the Wireless DataBox[®]. The sampling interval can be software selected from 2 hours to 48 hours based on the duration of the analysis.

Data transmission can take place even if the Wireless DataBox[™] is not connected to a PC (provided it is connected to a power socket). Each Wireless DataBox[™] can be connected up to 50 sensors.

For BOD, the analysis takes 5 to 30 days. For BMP, the analysis requires a minimum of 21 days (Italian UNI / TS standard) up to a maximum of 60 or 100 days, depending on the degradability of the substrate.

RESPIROMETRIC Sensor is powered by 2 lithium micro batteries that make the system free from any electrical or hydraulic connection.

The following VELP Scientifica refrigerated thermostats are available for controlled temperature incubation:

MODEL	CAPACITY		TEMPERATURE [°C]
	RESPIROMETRIC SENSOR SYSTEM 6 MAXI	RESPIROMETRIC SENSOR SYSTEM 6	
FTC 120	1	3	20, constant
FOC 120E CONNECT	1	3	3 + 50, programmable
FOC 120I CONNECT	1	3	
FOC 200E CONNECT	2	4	
FOC 200I CONNECT	1	4	
FOC 200IL CONNECT	1	4	

Table 1

1.1 Included items

Check the integrity of the stirring system upon receipt. The table below indicates the elements contained in the system packaging. Depending on the system model ordered, check the presence of all the following components, individually packaged, indicated in the column:

	BOD SENSOR SYSTEM 6	BOD SENSOR SYSTEM 10	RESP. SENSOR SYSTEM 6	RESP. SENSOR SYSTEM 6	RESP. SENSOR SYSTEM 6 MAXI	RESP. SENSOR SYSTEM 6 MAXI
PRODUCT CODE	SA10200136 SA10210136	SA10200137 SA10210137	SA10200146 SA10210146	SB10200146 SB10210146	SA10200156 SA10210156	SB10200156 SB10210156
SENSOR ¹	F102B0133	F102B0133	F102B0153	F102B0153	F102B0153	F102B0153
STIRRING STATION	F10200270 F10210270	F10200280 F10210280	F10200270 F10210270	F10200270 F10210270	F10200275 F10210275	F10200275 F10210275
WIRELESS DATABOX ²	-	-	A00000402	A00000402	A00000402	A00000402
TREBLE HOOKS MAGNETS	A00001056 N°6 pz	A00001056 N°10 pz	A00001056 N°6 pz	A00001056 N°6 pz	A00001056 N°6 pz	A00001056 N°6 pz
DARK GLASS BOTTLES	10001107 N°6 pz	10001107 N°10 pz	10001107 N°6 pz	10001107 N°6 pz	-	-
ALKALI CONTAINERS	10000904 N°6 pz	10000904 N°10 pz	10000904 N°6 pz	10000904 N°6 pz	10000904 N°6 pz	10000904 N°6 pz
GLASS BOTTLE 1000 ML	-	-	-	-	10008454 N° 6 pcs	10008454 N° 6 pcs
GL45 REDUCER WITH KOH CONTAINER	-	-	-	-	10005732 N° 6 pcs	10005732 N° 6 pcs
N° 1 Power cable						
N° 1 BOD Sensor identification label						

Table 2

- 1) The sensor box contains the RESPIROMETRIC Sensor or BOD Sensor with its instruction manual.
- 2) The receiver box contains the RESPIROSoft™ and its instruction manual.

NOTE

- Keep the entire shockproof packaging of the digestion system as it may be necessary for any future transport.

2. Instrument description



The RESPIROMETRIC Sensor System allow 6 bottles to be stored in small spaces. The recessed side handles allow easy insertion and extraction of the device from the refrigerated thermostat even with the bottles already inserted. The type of motor used for the stirring allows one to obtain continuous operation without risk of interruption or overheating which could compromise the test results.

The motion transmission takes place by means of a belt with a series of pulleys specially designed for this application.

3. Assembly and installation

Check that the power supply voltage of the device is the same as the operating voltage. Position the product so that the plug is always easily accessible.

With all VELP Scientifica refrigerated thermostats it is possible to power the RESPIROMETRIC Sensor System 6 stirring system by means of an external switch located on the front panel of the refrigerated thermostat.

4. Operation checks

To check that all the stirring points are operational, after having electrically connected the equipment, place a bottle on each stirring point and visually check that the stir bar, previously inserted, is moving.

5. End of work operations

At the end of each test it is recommended to not leave the instrument connected to the power socket. Clean the instrument with a cloth moistened with non-aggressive detergents after turning off the instrument.

When the system has not been used for some time, it is preferable to not leave the RESPIROMETRIC Sensors screwed onto the bottles so as not to unnecessarily stress the lip of the alkali container which acts as a seal. It is also recommended to remove the power supply batteries.

To wash the bottles after each test see the next chapter Maintenance.

6. Maintenance

Ordinary and extraordinary maintenance of the instrument is not required except for periodic cleaning of the instrument to be carried out as described in the previous paragraph.

After each determination, the bottles, the magnetic stir bars and the alkali containers must be washed carefully with hot water, possibly with the help of a pipe cleaner. It is not recommended to use detergents as these products have very high BOD values, so even small residual traces can alter the result of the measurements. Also the use of a chromo-sulphuric mixture is not recommended due to the high toxicity of chromium for microorganisms and the difficulty with which residual traces can be removed with a simple rinsing. It is the user's responsibility to decontaminate the unit in the event that hazardous substances remain on the surface or inside the device. If there is a doubt about the compatibility of a cleaning or decontamination product, contact the manufacturer or distributor.

7. Disposing of the unit



For definitive decommissioning, the instrument falls into the category of electrical and electronic equipment and therefore must be subjected to separate collection and cannot be disposed of as urban waste, according to the provisions of Directive 2002/96/EC. For more detailed information on recycling this product, please contact your municipal office.

8. Optional accessories

DESCRIPTION	CODE
Sensor Check	A00000135
Control test pads	A00000136
Complete bottle with two necks for BMP	A00000410

Table 3

9. Spare parts

DESCRIPTION	CODE
BOD Sensor	F102B0133
RESPIROMETRIC Sensor	F102B0153
Magnetic stir bar 6x35 mm	A00001056
Container for alkali	10000904
Dark glass bottle 500 ml	10001107
Glass bottle 1000 ml	10008454
GL45 reducer with KOH container	10005732

Table 4

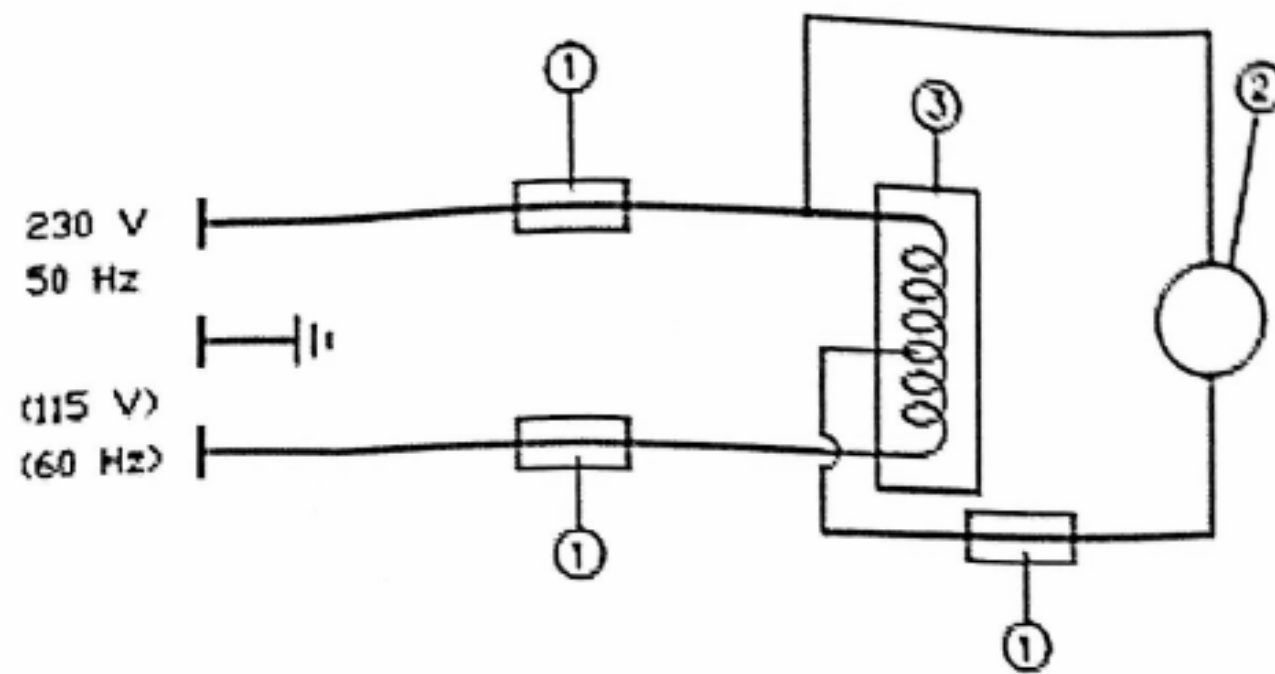
The spare parts mentioned above can be easily replaced by the operator. For other spare parts (and the complete list) contact the nearest VELP service center.

10. Technical data

	BOD SENSOR AND RESP. SENSOR SYSTEM 6	BOD SENSOR SYSTEM 10	RESP. SENSOR SYSTEM 6 MAXI
STIRRING PLACES	6	10	6
DIMENSIONS (BXHXP) MM	270x300x185	432x300x165	432x300x165
WEIGHT KG	1	3	3
POWER SUPPLY	230 V $\pm 10\%$ 50 Hz or 60 Hz or 115 V $\pm 10\%$ 60 Hz		
POWER	2 W		
STIRRING SPEED	60 rpm		
SAFETY CLASS	I IEC/EN61010-1		
MAX. ALTITUDE	2000 m		
MAX. HUMIDITY	85 %		
OVERVOLTAGE CATEGORY	II		
POLLUTION DEGREE	2		
MAX. NOISE LEVEL	52 dBA		
FUSES	250 mA 5x20 mm (230V) 500 mA 5x20 mm (115V)		
ADMITTED ROOM TEMPERATURE	Storage $-25 + +65^{\circ}\text{C}$ Operation $+5 + +40^{\circ}\text{C}$		

Table 5

11. Electric scheme



- 1) 5x20 mm delayed fuse 63 mA
- 2) Motor
- 3) Transformer

12. Analytical methods

12.1 Introduction

VELP RESPIROMETRIC Sensor System offers an easy-to-use method for monitoring and quantifying the gases produced during the anaerobic or aerobic degradation of organic material; processes widely applied both in the treatment of waste water and, more generally, for the production of renewable energy in the form of biogas.

The biochemical oxygen demand is an index of the amount of biodegradable organic substance present in a water sample.

During the oxidative degradation of the organic substance the aerobic microorganisms that are responsible for it consume oxygen, which is present as a gas dissolved in the liquid. The biochemical oxygen demand is expressed in weight of oxygen consumed per volume of liquid during a given period of time at a given temperature, for example $\text{mg O}_2 / \text{l}$ (or ppm = parts per million) in 5 days at 20°C .

In Italy the English abbreviation BOD (Biological Oxygen Demand) is used, in France the corresponding abbreviation is DBO (Demande Biochimique en Oxygène), in Germany BSB (Biologischer Sauerstoff Bedarf).

The completion of biological oxidation at 20°C requires a period of time between 21 and 28 days (BOD ultimate, BOD21, BOD28).

Since these periods are very long compared to practical needs, conventionally the measurement is carried out after 5 days of incubation during which, at 20°C , a weight of oxygen equal to about 70% of what would happen in 21-28 days is consumed.

Compared to the chemical method for measuring the oxygen demand (COD = Chemical Oxygen Demand), the relationship $\text{COD} = \text{BOD}_{21}$ exists only in the case in which the organic substances present are completely biodegradable, as in the case of glucose. However, many natural and synthetic organic compounds present in water are not biodegradable (lignin) or are very slowly compared to the conditions of the assay (cellulose) while they are rapidly oxidized under the conditions of the chemical method (acid solution of hot dichromate). It follows that for the most part the COD value is higher than that of the BOD without a definable a priori relationship except in the case of distinctive waste water.

Being a process carried out by living organisms, the consumption of oxygen is affected by the temperature: it can be detected roughly between 0°C and 50°C depending on the microorganisms present and the speed with which it increases with the increase of the medium's temperature.

Standard and official methods require that the measurement be conducted at 20°C with a clear reference to the average conditions of the natural environment.

However, for particular application needs (hot or cold areas) it can be carried out at temperatures other than 20°C .

The BOD measurement is extensively used to evaluate the load of organic substance to which a biological treatment plant for waste water is subjected or its purification effectiveness, but it can be used for the same evaluation in the case of surface bodies of water (rivers, lakes, seas).

The assay is conducted in the dark to avoid interference due to the production of photosynthetic oxygen by microalgae.

The measurement of BMP (Biochemical Methane Potential) is a technique used to determine the methane potential or the anaerobic biodegradability of substrates such as sludge, waste water and more generally biomass waste. Each test involves the preparation of both "test bottles", in which a substrate is mixed with an anaerobic bacterial culture that is normally taken from an active digester, and "blank bottles" into which the microbial inocula is dosed in the absence of substrate. The bottles are then stored at a stable temperature (e.g. 35°) and constantly mixed for a period of time to ensure complete anaerobic degradation of the substrate. During the test period, the biogas consisting mainly of methane and carbon dioxide is generated as a product of the anaerobic degradation of the organic content of the substrate. Since carbon dioxide is an acid gas, it can be removed by alkaline adsorption, thus allowing the measurement of the methane generated by the substrate; the BMP is then calculated as the difference between the total volume of methane produced by the "test bottle" and that accumulated in the "blank bottle", thus referring the result to the mass of volatile solids or COD (chemical oxygen demand) of the substrate added to the test bottles. If

the COD of the substrate is known, it is also possible to determine the anaerobic biodegradability of the substrate itself by dividing the volume of methane produced on a COD basis by the theoretical volume of methane that can be defined, under standard temperature and pressure conditions (STP) by the chemical equivalence for which 1 g COD corresponds to 0.35 ml of CH₄.

BMP tests are used to characterize a wide variety of substrates, both of the liquid type (primary and secondary sludges from sewage treatment plants, animal waste, wastewater or waste by-products from the food industries, ...) and of the solid type (organic waste matter of municipal solid waste, lignocellulosic matrices, ...). Since the mixing conditions are essential for a successful experiment, in the case of solid substrates it is often preferable to proceed with a pre-treatment to reduce the size of the substrate in order to not only facilitate mixing, but also to improve the representativeness of the sample tested.

12.2 Respirometric measurement methods

The solubility of oxygen in water is limited. In distilled water at sea level (760 mm Hg) the maximum (100% saturation) at 20°C is 8.84 ppm in equilibrium with humid air, whose oxygen content by volume is 20.95% (23.5% in weight).

Solubility decreases with increasing altitude (lower pressure) and salt content. BOD can be measured by determining the dissolved oxygen content (by titration or with a specific electrode) in a water sample at the beginning and at the end of the incubation period.

A measurable level of oxygen must still be present at the end of the incubation; a zero value is meaningless.

Starting from an initial value close to saturation, values lower than 8 mg / l of oxygen remain available, i.e. measurable BOD. There are sewage whose BOD can have values of 100000 ÷ 150000 ppm, in most cases hundreds or thousands of ppm must be detected.

Consequently, to use the dissolved oxygen determination it is necessary to strongly dilute the test samples and prepare a series of different dilutions in order not to have all the samples diluted to zero oxygen content at the end of the incubation. Incubation takes place in unshaken bottles and each sample cannot be used further after being used for titration of oxygen content.

There are methods that provide for a continuous supply of oxygen dissolved in the incubated sample. In one case, to keep the content constant the oxygen to be supplied is produced by electrolysis and the amount of electricity consumed for its production is measured over time and converted into the amount of oxygen consumed. With a simpler method, oxygen consumption is measured manometrically, according to the principle of Warburg's respirometer, widely used in biochemistry. Samples with BOD below 800-900 ppm should not be diluted.

Samples with very high BOD undergo a much lower dilution than that required by the titration method and this makes the operating conditions much more similar to those of the plant or natural environment to be monitored. The sample is stirred continuously during the incubation and this avoids the formation of concentration gradients.

The supply of oxygen is continuous and facilitated by the shaking, providing the sample with average concentrations far from potential extreme values (100% saturation or anoxia).

The measurement of BMP (Biochemical Methane Potential) represents the corresponding anaerobic tests of the BOD (Biochemical Oxygen Demand), in fact it takes place by putting the organic test substance in contact with an adequate microbial inocula suitable to degrade the test substance anaerobically. During the test, the production of biogas and / or methane is monitored over time, which is proportional to the quantity of organic substance gradually degraded.

The result is usually expressed in terms of mlCH₄ @ STP / gVS, which represents the volume of methane produced per unit of organic substance quantified in terms of volatile solids.

The tests are carried out in batches by putting a known quantity of the substance of interest, the test substance, into contact with an inocula of anaerobic microorganisms in a suitable reactor. The reactor is normally constituted by a simple glass bottle, sealed and kept under controlled and constant conditions of temperature and stirring. The bottle is initially brought under anaerobic conditions thanks to the flushing of an anoxic gas mixture (typically N₂ or N₂/CO₂). With the test thus started, as the degradation process proceeds, biogas will be produced in the bottle which can be removed and measured volumetrically or, more simply, quantified in terms of overpressure generated, if the gas is kept inside the reactor (manometric method).

For the BMP test, temperature is of great importance as it affects both biological activity and equilibrium and chemical-physical transformations. It also affects the biogas measurements and must therefore be carefully kept constant. The temperature value at which to incubate the test bottles depends on the type of inocula sludge chosen. Normally, the tests are carried out in at a mesophile temperature of 35 ° - 40°C, or at a thermophile temperature of 50-55°C.

VELP RESPIROMETRIC Sensor System can also be used for further tests:

- Determination of the aerobic biodegradability of plastic materials in aqueous medium
- Soil quality: laboratory methods for the determination of soil microbial respiration
- Biodegradation of Monomers
- Assessment of the effects of disinfection on BOD determination
- Denitrification activity of the biomass

13. BOD analysis

13.1 Reagents required

For carbon dioxide absorption:

- Caustic Potash (KOH) in flakes, commercial grade (a particularly pure type is not necessary).

or

- Non-deliquescent soda lime, 1.0-1.7 mm granules.

For possible neutralization of samples before determination:

- Sulfuric acid (H_2SO_4) 1.0 N solution.
- Sodium hydroxide (NaOH) 1.0 N solution.

For the possible addition of nutrients:

- High quality distilled water free of copper and organic matter
- Iron chloride ferric hexahydrate ($\text{FeCl}_3 \cdot 6 \text{H}_2\text{O}$)
- Anhydrous calcium chloride (CaCl_2)
- Magnesium sulphate heptahydrate ($\text{MgSO}_4 \cdot 7 \text{H}_2\text{O}$)
- Diacid potassium orthophosphate (KH_2PO_4)
- Potassium orthophosphate monoacid (K_2HPO_4)
- Sodium orthophosphate monoacid ($\text{Na}_2\text{HPO}_4 \cdot 7 \text{H}_2\text{O}$)
- Ammonium chloride (NH_4Cl).

For the possible dechlorination of the samples:

- Sodium sulfite (Na_2SO_3).

For the possible inhibition of nitrification:

- Allylthiourea (Thiosinaminum) $\text{C}_4\text{H}_8\text{N}_2\text{S}$.

or

- 2-chloro-6 (trichloromethyl) pyridine, $\text{C}_6\text{H}_3\text{Cl}_4\text{N}$ (CTCMP, N-Serve).

Additive for mineral fertilizers aimed to lower nitrogen losses in soils.

13.2 Choice of scales

The equipment is used to determine BOD values below 90-250-600-1000-4000 ppm respectively. If information is available on the presumable BOD value of the samples under examination, the scale to be used can be appropriately chosen.

For samples with BOD higher than 1000 ppm it is necessary to dilute them appropriately before the test, possibly using subsequent dilutions to maintain significance to the volume as it is sampled, especially in the presence of turbid or viscous solutions.

For example: 50 ml of sample as it is measured with a 50 ml graduated cylinder, brought to 500 ml in a 500 ml graduated cylinder with high quality distilled water (distilled on permanganate) and added with nutrients (point 13.13): dilution obtained 1:10.

For further dilutions operate in the same way making sure to homogenize the diluted sample by shaking it with a clean glass (or plastic) rod.
Multiply the final result by the dilution factor.

In the event that there is no prior information on the BOD of a sample under examination, a series of three bottles must be prepared respectively containing the undiluted sample; diluted 1:10; diluted 1: 100 and a choice of scales based on the highest known value for samples belonging to the same type as the one in question (livestock sewage, discharges from particular industries, etc., point 13.6).

SCALE	VOLUME OF SAMPLE
0 + 90 mg O ₂ /l	400 ml
0 + 250 mg O ₂ /l	250 ml
0 + 600 mg O ₂ /l	150 ml
0 + 1000 mg O ₂ /l	100 ml
0 + 4000 mg O ₂ /l	25 ml
PRE	Refer to the method

Table 6

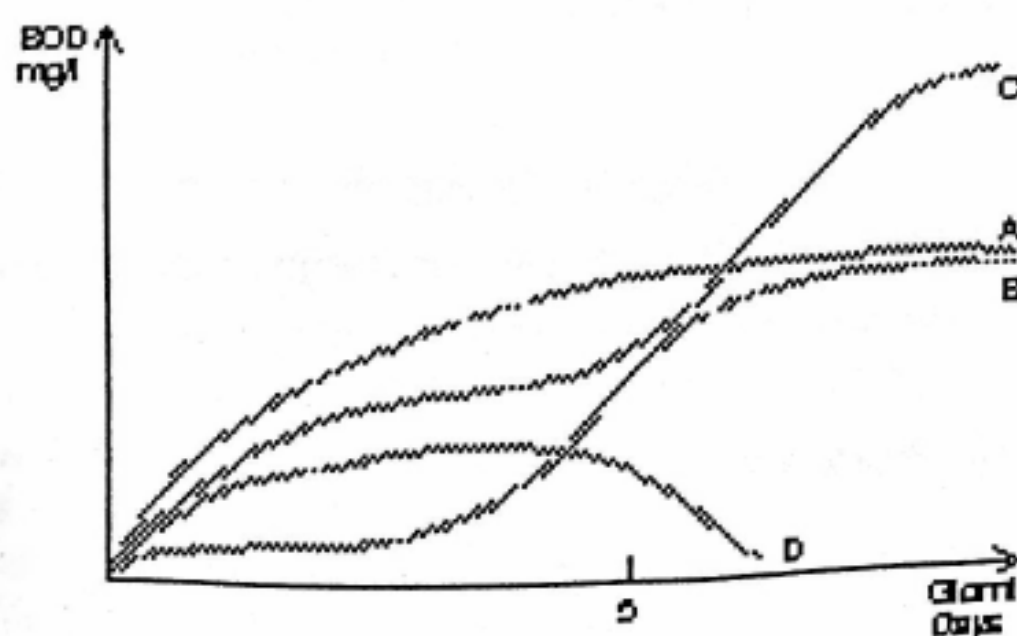
13.3 Examples of experimental curves

To make sure the aerobic degradation of substances in the samples under examination proceed normally, the BOD values are noted daily and are graphed. Note the following criteria:

- 1) The amount of oxygen consumed daily decreases progressively compared to that consumed the previous day.
- 2) The cumulative amount of oxygen consumed grows progressively up to almost be stabilized at a constant value.

The following curves highlight particular situations:

- A) Typical degradation curve of carbonaceous substances in the presence of suitable bacterial flora and adequate amounts of nutrients.
- B) A trend due to the initial presence of a low number of adapted bacteria to the substrate or substances inhibiting bacterial growth.
- C) When oxygen consumption starts to rise again, typically on the 6th-7th day from the beginning, it is in the presence of nitrifying bacteria that oxidize the reduced compounds nitrogen (ammonia) to nitrites. This effect can be eliminated using substances inhibiting nitrification (Section 12.11).
- D) Occurs when there are leaks in the system seals.



13.4 Indicative values of BOD⁵₂₀

BOD⁵₂₀ indicates values obtained after 5 days of incubation at 20°C.

The following values can serve as a guide for choosing the scales and dilutions to be made. The estimate can also be made by knowing the quantity by weight produced, for example 65 g BOD / population / day, and the corresponding water consumption, for example 150 litres / population / day, corresponding to approximately 450 mg BOD⁵₂₀/l, or 400 l / population / day equal to about 160 mg BOD⁵₂₀/l.

UNTREATED WASTE	BOD MG/L
Paper mills	200-4000
Tanneries	500-1500
Distilleries	12000-35000
Oil mills	15000-70000
Dairies	200-6000
Slaughterhouses	1000-2500
Textile-dyeing	1000-12000
Raw pigsty sewage	7000-9000
Pharmaceutical industry	800-10000
Domestic waters	800-2000

Table 7

13.5 Sampling

Collect the sample in a clean tank. If the BOD analysis is performed within 2 hours, there is no need to refrigerate the sample. Otherwise, store it at a temperature below 4°C. Do not perform BOD analyses on samples stored for more than 24 hours.

In evaluating the BOD on the sample as it is, it may happen that the suspended material is not well mixed and therefore the oxygen in the bottle is consumed faster compared to another one. It is good practice to homogenize the sample with a magnetic stirrer to obtain repeatable results, avoiding damage to microorganisms.

13.6 Pre-treatments of samples

In the following cases, the water sample must be subjected to appropriate pretreatments before being used for the determination of the BOD:

- if the pH value does not fall within the range of 6.5-7.5.
- if it concerns water disinfected with chlorine, chlorine dioxide, ozone, etc.
- if you are in the presence of toxic substances for bacteria.
- if the water is rich in nitrifying bacteria.
- if the bacterial population is scarce.
- if the content of nutritional elements (nitrogen, phosphorus, microelements) is insufficient.

Many of these cases refer to industrial wastewaters.

13.7 Neutralization of samples

The bacterial flora has an optimal pH range between 6.5 and 7.5. Outside of this scope, the BOD results are lower than they should be. It is therefore necessary to correct the pH using 1.0 N solutions of sodium hydrate or sulfuric acid, respectively. The volume of acid or base added must not exceed 0.5% of the volume of the test sample. If necessary, it is advisable to use more concentrated solutions.

13.8 Removal of free and combined chlorine

Disinfectants containing chlorine are normally used for the elimination of bacteria.

In case you want to determine the BOD value of waters containing disinfectants, it is necessary to eliminate free or combined chlorine by adding small volumes of a sodium sulphite solution about 0.025 N (1.58 g / l in distilled water).

This solution is not stable and must be used within the day.

Such samples require microbial inocula after disinfectant removal.

13.9 Removal of toxic metals

Some industrial effluents (e.g. galvanic) contain toxic metals. Bringing the pH of these effluents to 8.5 results in the precipitation of most of the metals such as hydroxides or oxides which can be removed by filtration. The pH is then brought back between 6.5 and 7.5 and the sample is inoculated with bacteria. In some cases, a reduction in toxicity within reasonable limits can be achieved simply by diluting the sample. However, the problem of sample toxicity requires specific investigations.

13.10 Inhibition of nitrification

In some cases, it is necessary to exclude from the determination of the BOD the contribution attributable to nitrification, for example in the case of effluents that have undergone biological treatment, effluents inoculated with biologically treated effluents, river waters.

Nitrification normally begins after 5 days (at 20°C), therefore its influence on the BOD measurement is negligible. If nitrifying bacteria are present in large numbers in the test sample, their activity can be inhibited by adding special chemicals.

According to the methods for water analysis (IRSA-CNR Method E-008, 1981) this is obtained by adding the following volumes of a 0.05% solution (50 mg in 100 ml of water) of allylthiourea (ATU) to each BOD bottle:

VOLUME OF SAMPLE	VOLUME OF INHIBITING SOLUTION
ml 100	ml 0.3
ml 150	ml 0.5
ml 250	ml 0.8
ml 400	ml 1.3

Table 8

Store the allylthiourea solution at 4°C in a dark environment. The solution is stable for at least 2 weeks.

According to American standard methods, the same purpose can be achieved by adding the volumes indicated above of a 0.35% solution (350 mg in 100 ml of water) of 2-chloro-6 (trichloromethyl) pyridine (TCMP). If a nitrification inhibitor was used this should be indicated in the presentation of results.

13.11 Microbial inocula

If the test sample does not contain a sufficient number of microorganisms, for example when examining samples of industrial waste water or obtaining curves of type B described in point 12.5, it is necessary to inoculate the test sample with suitable bacteria to initiate the process.

Generally, a portion of water taken a few kilometres downstream of the discharge of the water to be analysed is used as inocula, alternatively it is possible to use river or lake water containing urban waste. Today there are ready-to-use bacterial inocula for BOD analysis on the market.

Normally an inocula volume equal to 10% of the sample volume is used.

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The BOD of the supernatant of an urban sewerage water can be between 50 and 300 ppm with values varying over 24 hours. The BOD of the inocula (blank) must be determined at the same time as that of the inoculated samples by diluting it with high quality distilled water in the same ratio used to inoculate the samples (example 100 ml / l).

The value obtained for the inocula must be subtracted from the values obtained for samples inoculated with the same. The trigger for oxygen consumption depends on the effectiveness of the inocula. In some cases it is advisable to use increasing volumes of inocula, for example 100-200-300 ml per litre sample. At the end of the incubation, the value of the inoculated sample which will have recorded the highest BOD after the subtraction of the BOD value to be attributed to the volume of inocula used will be considered valid.

13.12 Addition of nutrients

Industrial waste water has an insufficient quantity of nutritional elements (nitrogen, phosphorus, microelements) which are essential for the regular development of a bacterial population. For this reason, it is often necessary to add nutrients using the following solutions:

- **Solution A** – dissolve 0.25 g of ferric chloride exahydrate ($\text{FeCl}_3 \cdot 6 \text{H}_2\text{O}$) in 1 l of distilled water.
- **Solution B** - dissolve 27.5 g of anhydrous calcium chloride (CaCl_2) in 1 l of distilled water.
- **Solution C** - dissolve 22.5 g of magnesium sulfate heptahydrate ($\text{MgSO}_4 \cdot 7 \text{H}_2\text{O}$) in 1 l of distilled water.
- **Solution D (buffer)** - dissolve 8.5 g of monobasic potassium phosphate (KH_2PO_4), 33.4 g of disodium phosphate heptahydrate ($\text{Na}_2\text{HPO}_4 \cdot 7 \text{H}_2\text{O}$), 21.7 g of dipotassium phosphate (K_2HPO_4), 1.7 g of ammonium chloride, ($\text{NH}_4 \text{Cl}$) in about 500 ml of distilled water. Make up to 1000 ml and mix.
- **Final pH = 7.2.** Samples without nutrients with a B.O.D. lower than 800-900 ppm are added with 1 ml of each solution (A-B-C-D) for each liter of sample.

The pH of the dilution water must be between 6.8 and 8.

For nutrient-free samples with BOD below 800-900 ppm, it is recommended to add 1 mL of each solution (ABCD) to 1 L of distilled water. This solution is used for the preparation of the inocula and standard solutions.

13.13 Dilution water

For samples to be diluted, it is advisable to prepare distilled water for dilution with the addition of nutrients: for each litre of distilled water, 1 ml of each of the solutions (A-B-C-D) will be added.

This dilution water should not have a BOD5 greater than 0.2 ppm.

If high dilution factors (1: 100, 1: 1000) are used and you are not sure of the quality of the distilled water used to dilute, it is also advisable to determine the BOD of the latter which will be subtracted from that of the diluted sample, before multiplying the result by the dilution factor.

13.14 Control standard

The precision of method can be evaluated by using solutions with a known content of D-glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and L-glutamic acid ($\text{C}_5\text{H}_9\text{NO}_4$), pure grade and previously dried at $(105 \pm 5)^\circ\text{C}$ for 1 hour. Dissolve (150 ± 1) mg of both reagents in 1000 ml of distilled water.

- Weigh (150 ± 1) mg of both
- Dissolve in water, make up to 1000 ml with dilution water with the A-B-C-D solutions and mix.
- Add 1 ml of inocula to the solution.

The theoretical BOD₅ is equal to 210 ± 20 mg O₂ / l (according to the APAT-IRSA.CNR 5120 "Biochemical oxygen demand - BOD₅ method) and the BOD₇ is 225 ± 20 mg O₂ / l.

Similarly, a 50 mg / L sodium acetate solution can be prepared.

Introduce 400 ml of this solution into the bottles and analyse with a 90 ppm scale.

The theoretical BOD₅ is equal to 31 ± 5 mg O₂ / l.

To check the operation of the system it is possible to use Control Test Tablets (code A00000136) with the 600 scale, which is able to simulate oxygen demand similar to the BOD measurement method. The introduction of a tablet makes it possible to measure the amount of oxygen that is chemically absorbed by the system.

13.15 BOD calculation example

250 ml of sample diluted 1:10 with distilled water are introduced into a bottle, inoculated with 1 or 2 ml of activated sludge.

After 5 days, a BOD of 200 mg / l is measured.

In another bottle, the same amount of distilled water and inocula is introduced and, after 5 days, a BOD of 15 mg / l (blank) is detected.

The real BOD, taking into account the dilution carried out on the sample, will be:

$$\text{real BOD} = (200-15) \times 10 = 1850 \text{ mg/l}$$

Collecting data table:

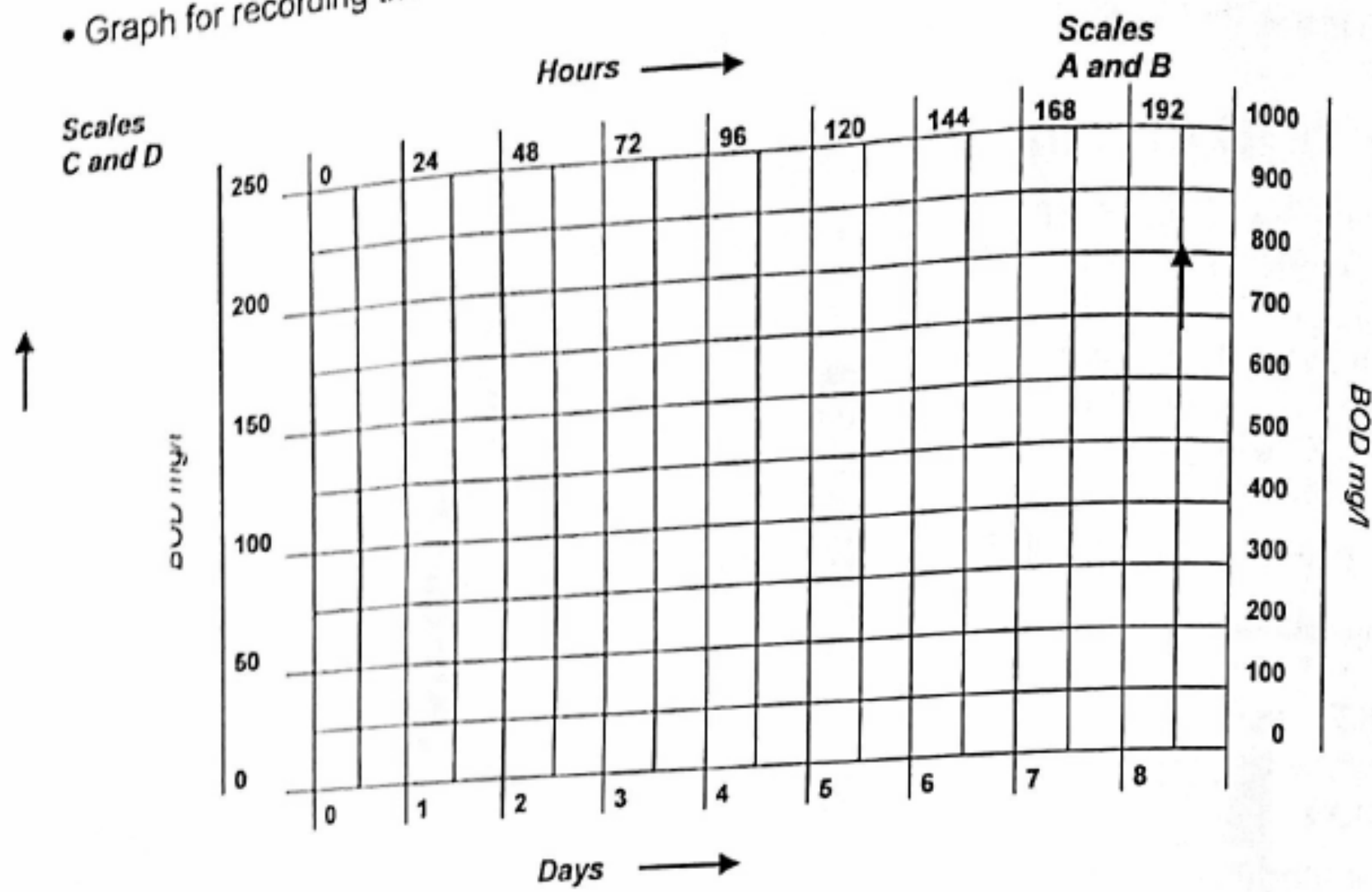
SAMPLE

POSITION	1	2	3	4	5	6
N° SENSOR						
SCALE						
VOLUME						
DILUTION						
BOD _N VALUE						

By recording the BOD values on the following graph, it is possible to construct the BOD curve.

In wireless mode, this curve will be created automatically by the software.

- Date:
- Hour of incubation start:
- Operator:
- Graph for recording the BOD curve:



14. BMP analysis and anaerobic tests

14.1 Reagents required

For the absorption of carbon dioxide:

- Caustic Potash (KOH) in flakes, commercial grade (a particularly pure type is not necessary).

or:

- Non-deliquescent soda lime, 1.0-1.7 mm granules.

To correct the pH of the mixture (inocula, substrate, water and nutrients):

- Hydrochloric acid (HCl), 1 N solution.
- Sodium hydroxide (NaOH), 1 N solution, or sodium bicarbonate (NaHCO₃), powder.

To correct the pH of the inocula:

- Sodium bicarbonate (NaHCO₃), powder.

Solutions of nutrients and trace elements (optional):

- Distilled water
- Anhydrous potassium dihydrogen phosphate (KH₂PO₄).
- Disodium hydrogen phosphate dodecahydrate Na₂HPO₄ • 12H₂O,
- Ammonium chloride (NH₄Cl),
- Calcium chloride dihydrate (CaCl₂ • 2H₂O),
- Magnesium chloride hexahydrate (MgCl₂ • 6H₂O),
- Iron (II) chloride tetrahydrate FeCl₂ • 4H₂O,
- Manganese chloride tetrahydrate (MnCl₂ • 4H₂O)
- Boric acid (H₃BO₃)
- Zinc chloride (ZnCl₂)
- Copper (II) chloride (CuCl₂)
- Disodium molybdate dihydrate (Na₂MoO₄ • 2H₂O)
- Cobalt chloride hexahydrate (CoCl₂ • 6H₂O)
- Nickel chloride hexahydrate (NiCl₂ • 6H₂O)
- Disodium selenite (Na₂SeO₃)

Deoxygenated gas (e.g. nitrogen or argon).

14.2 Examples of experimental curves

Here are some qualitative trends of pressure curves that can be obtained in the BMP tests and their probable interpretation.

A) **Pressure produced by the consumption of an organic substance that has both rapidly and slowly degradable components.** The components that are more easily degradable (and normally present in solute form) are initially consumed and converted to methane. Subsequently, the degradation rate is linked to the speed with which the first phases of the anaerobic degradation process proceed, i.e. disintegration and hydrolysis. The speed with which these phases proceed is proportional to the concentration of the substance to be hydrolysed, and therefore decreases during the test, i.e. as the organic substance to be hydrolysed is consumed. The observed biogas production trend in this case can be well described by an exponential curve which reaches an asymptotic value at the end of the test. When the rapidly degradable component is predominant, a rapid production of biogas is

observed in the initial test phase, with an almost rectilinear pressure trend associated with it, this trend then tends to break due to the composition of the less rapidly degradable component.

B) Pressure produced by the consumption of an organic substance with slow degradability, which must undergo the hydrolysis process. This trend is typical for substrates such as microcrystalline cellulose, for which a short (few hours) delay phase or initial lag is normally observed.

C) Pressure produced by the consumption of an organic substance which requires an adaptation phase of the biomass, known as the lag phase, before the latter develops the ability to degrade the test substance. This lag phase normally has a longer duration than in the previous case.

D) Pressure produced by the consumption of a rapidly fermentable organic substance which leads to a rapid development of the acidogenesis phase, releasing carbon dioxide and hydrogen into the headspace. The subsequent phase of hydrogenotrophic methanogenesis, which consumes these two gases to give methane, can lead to a temporary reduction in the pressure in the bottle, which is not attributable to gas leaks.

E) Case in which the test is compromised by a loss of pressure which requires the repetition of the test itself after a check of the seals (gaskets and separators).

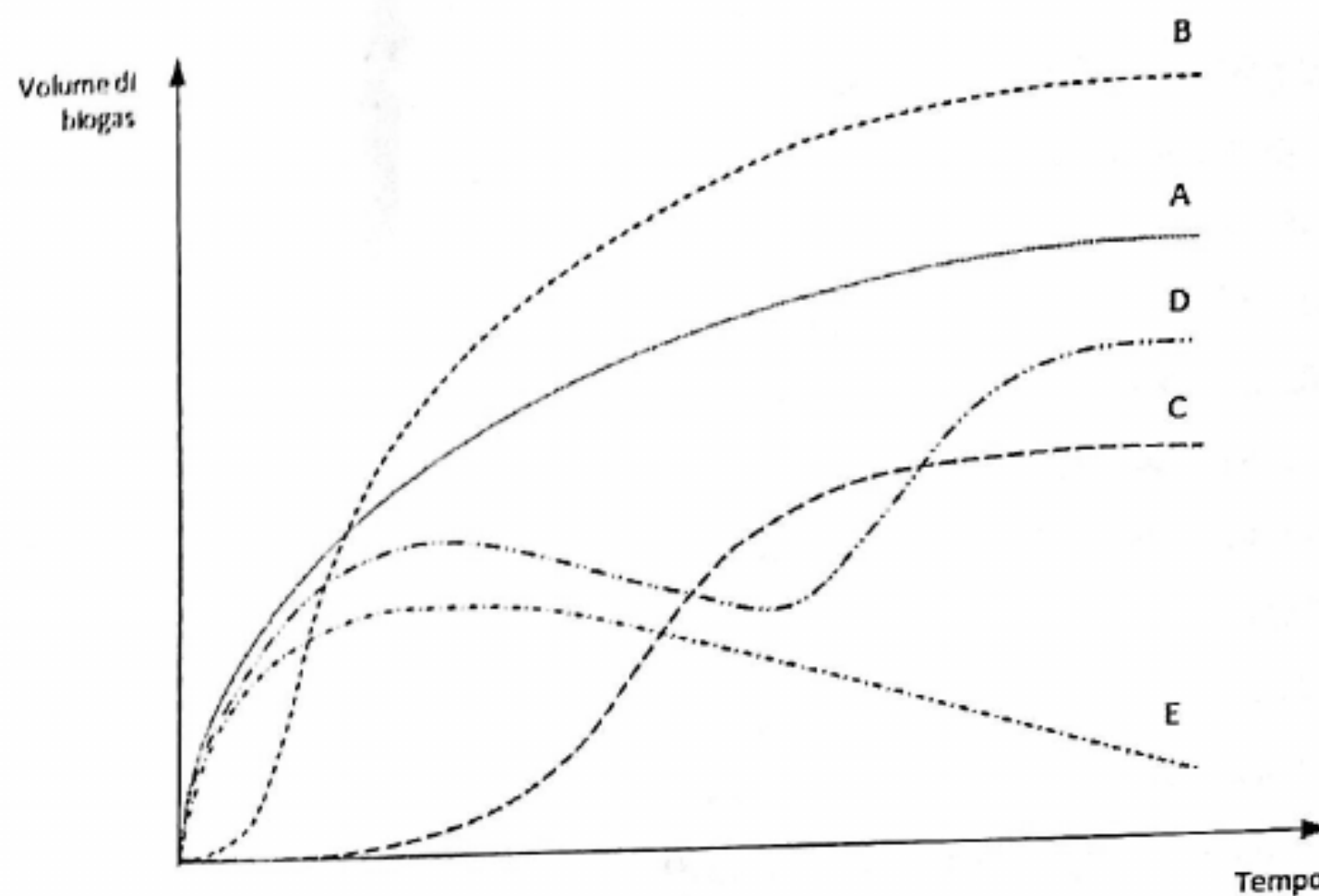


Figura 1. Andamenti qualitativi di curve di pressione da prove di BMP

14.3 Indicative values of BMP

UNTREATED WASTE	VALUE RANGE FOR BMP [NMLCH ₄ / GVS]
Pig sludge	150 - 350
Bovine sludge	230 - 250
Corn silage	290 - 370
Feed silage	270 - 370
Hay	270 - 290
Whey	360 - 480
Vegetable oil	560 - 730
Municipal sewage plant sludge	70 - 450
Meat processing industry waste	300 - 920
Fish processing industry waste	40 - 350

Table 9

14.4 Sampling

The test substrate must be collected in a clean container, stored at a temperature below 4°C, and promptly characterized, at a minimum, in terms of total dry matter and organic content (COD or Volatile Solids); in order to correctly set up the tests and subsequently report the methane production obtained to the organic substance content initially present.

As for the inocula, it is always required that it be taken from a digester operating at the same temperature at which the BMP will be performed. Once collected, it must be used within a short time (3-4 weeks) and stored under anaerobic conditions, avoiding excessive contact with oxygen. After sampling, a pre-incubation period of 5-7 days at the test temperature is recommended, not only to allow the biomass to re-acclimate to the temperature conditions at which the BMP test will be performed, but also in order to allow inocula to consume the organic matter already present in it. As for the substrates, the inocula must be also characterized, at a minimum, through the analyses of total and volatile solids in order to correctly set up the test.

14.5 Sample pretreatment

A typical pre-treatment aimed at homogenizing the sample consists in reducing the size: greater uniformity of the matrices helps to make the analysis more reproducible. The sample size can be reduced by manual cutting or mechanical trituration. The pre-treatment must not change the average composition of the sample, nor, if possible, the VS content. A relatively small size (in the order of a few mm) allows an increase in the specific surface of the substrate and facilitates contact between the matrix and bacteria, thus speeding up the degradation reactions. The size of the substrates therefore affects the speed with which the reactions take place, while there is currently no general confirmation in the literature that the size also has an influence on the ultimate achievable BMP value.

14.6 Neutralization of samples

A particularly important factor is the buffer capacity of the suspension tests, which must be proportional to the amount of acids expected to be obtained during the hydrolytic phase of the digestion process.

The optimal pH for biomethanization is between 7 and 8, trying to limit the variation during the test to no more than ± 0.2 pH units. It is therefore necessary to correct the pH of the final test mixture, if outside this range, using hydrochloric acid (HCl), in 1 N solution or sodium hydroxide (NaOH), in 1 N solution, or sodium bicarbonate (NaHCO_3). The aim is to ensure a sufficient amount of alkalinity to the process, which instead consumes alkalinity.

It is possible to adjust the pH of the anaerobic inocula in advance. The operation of adjusting the pH of the inocula, if performed before starting to dose the inocula itself in each of the bottles, allows one to improve the replicability between tests (the inocula sample used is more homogeneous in terms of alkalinity content). In fact, it often happens that the inocula has a low pH (7.1-7.2), which is an indication of low buffering power which may not be sufficient to guarantee enough alkalinity to the process. If possible, it is always advisable to check the alkalinity content of the inocula: if it is lower than 3 g CaCO_3 / L it is recommended to proceed by adding alkalinity in the form, for example, of sodium bicarbonate.

14.7 Microbial inocula

The inocula is responsible for providing the microbial consortium capable of initiating the process of anaerobic degradation of the test substance. The choice of the inocula is of fundamental importance for the success of the test and for the representativeness of the result. In general, if the biodegradability tests concern specific organic molecules, it is good that the inocula has already been exposed to the test substance, or to substances similar to it, in order to limit the acclimatization and stabilization period of the bacterial populations. When this is not possible, it is advisable to obtain sludge from different anaerobic digestion plants to have a well diversified population. It can be taken from an operational anaerobic digester; for example from a traditional mesophilic digester of the sludge line of a civil or cooperative wastewater purifier, a livestock manure digester, which could have better acclimatization to high ammonia concentrations, or again from an anaerobic digestion plant

of soluble industrial waste with granular biomass, which normally has excellent acidogenic and methanogenic activity.

It is also useful to provide a stabilization phase of the inocula, by incubation in endogenous conditions for a few days (usually 5-7), which guarantees the consumption of any organic residues that are not completely degraded.

In addition to the type and activity, the initial concentration of the inocula or, more precisely, the ratio between the test substance and the inocula, defined as the F / M ratio, and normally quantified in terms of volatile solids (grams of VS of test substance in relation to grams of VS of inocula). This ratio is in fact considered one of the test conditions that can significantly influence the results obtained as it expresses the initial organic load level to which the microorganisms are exposed. If poorly defined, an inhibition and a delay or incompleteness of degradation can develop. If the values of the hydrolysis and methanogenesis rates are not known, it is usually suggested to work with F / M values between 0.25 gVS / gVS and 1 gVS / gVS.

14.8 Addition of nutrients

The medium, or nutrient solution, serves to ensure the availability of nutrients (macro and micro-nutrients, trace elements, vitamins) necessary for the balanced development of the bacterial consortium. If the test is carried out on a single substance, the composition of the medium can be particularly complex; in the case of biodegradability tests of wastewater or organic waste, this formulation can be reduced to the contribution of mineral salts and, possibly, nitrogen.

The Italian regulation UNI / TS 11703 - 2018 suggests the adoption of the following nutrient solutions to be added, in each test bottle of BMP:

- Solutions A and B in the ratio of 5% of the final test volume (working volume)
- Solution C in the ratio of 1% of the final test volume (working volume)

Solution A - dissolve 2.7 g of anhydrous potassium dihydrogen phosphate (KH_2PO_4), 11.2 g disodium hydrogen phosphate dodecahydrate ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$) and 5.3 g of ammonium chloride in 0.5 l of distilled water, using a volumetric flask.

Solution B - dissolve 0.75 g of calcium chloride dihydrate ($\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$), 1.0 g of magnesium chloride hexahydrate ($\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) and 0.2 g of iron (II) chloride tetrahydrate ($\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$) in 0.5 l of distilled water, using a volumetric flask.

Solution C - dissolve 0.05 g of manganese chloride tetrahydrate ($\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$), 0.005 g of boric acid (H_3BO_3), 0.005 g of zinc chloride (ZnCl_2), 0.003 g of chloride (II) of copper (CuCl_2), 0.001 g of disodium molybdate dihydrate ($\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$), 0.1 g of cobalt chloride hexahydrate ($\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$), 0.01 g of nickel chloride hexahydrate ($\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$) and 0.005 g of disodium selenite (Na_2SeO_3) in 1 l of distilled water, using a volumetric flask.

14.9 Dilution water

In the case of substrates with a very high solids content, it may be necessary to add dilution water to the test bottles, in order to guarantee humid digestion conditions, thus limiting the total solids content in the test mixture. (<50 g / L). For each test bottle, it is recommended to contain the volatile solids concentration of the inocula and substrate, respectively between 5-10 gVS / L and 1-5 gVS / L. For the BMP test, it is preferable to use tap water and not deionized water to avoid damaging the bacterial biomass through unwanted osmosis processes.

14.10 Control standard

It is advisable to periodically carry out control tests of the instrumentation and of the procedures adopted by means of BMP tests on reference substrates such as microcrystalline cellulose and solutions of known sodium acetate. Although there are no internationally recognized criteria yet, the Italian standard UNI / TS 11703 - 2018 suggests the following acceptance ranges:

- Microcrystalline cellulose: $335 \pm 25\%$ NmLCH₄/gVS
- Cellulose acetate: $332 \pm 5\%$ NmLCH₄/gCOD

14.11 BMP calculation example

It is possible to convert the pressure measured into the volume of biogas (or methane) produced, using the following equation:

$$G_t = V (\Delta P_t) / P_0 \cdot 273,15 / (273,15 + T)$$

Where:

- G_t [NmL] is the volume of biogas produced at time t ;
- V [mL] is the volume of the headspace;
- ΔP_t [hPa] is the difference between the pressure measured at time t and the pressure at time zero.
- P_{standard} [hPa] pressure in the standard condition, 1013 hPa.
- T [°C] is the temperature of the biogas in the headspace (at which the pressure is read).

Once the pressures have been converted into volumes, the test blanks (VB) are averaged. The average production of blanks is then subtracted from the production of each test bottle (VS, i), so as to obtain the net production of methane which, once reported to the mass of VS (or COD) of the substrate dosed in each bottle (MS, i), determines the value of BMP (NmL / gVS or NmL / gCOD).

$$[BMP]_i = (V_{(S,i)} - V_B) / M_{(S,i)}$$

Where:

- BMP_i [NmL / gVS or NmL / gCOD] is the final BMP value of the i -th bottle;
- VS, i [NmL] is the volume of methane accumulated in the i -th test bottle under normal conditions of pressure and temperature;
- VB [NmL] is the average of the volumes of methane accumulated by the blank bottles under normal conditions of pressure and temperature;
- MS, i [gVS or gCOD] is the mass of substrate dosed in the i -th bottle.

Note that the VS content of inocula in the blank bottles should always be the same as in the test bottles. Otherwise, the production of the blank will be recalculated with respect to the ratio between the quantity of inocula VS present in the blank bottles and that which is present in the test bottles; only later will it be possible to subtract the production of blank from the production of test bottles.

15. Analytical procedure

BOD analysis

- Set the incubator temperature to 20.0°C
- Connect the Wireless DataBox TM and fill in the BODSoft™ software database
- Place 5-10 KOH pellets in the alkali collector, making sure they remain below the holes. If any KOH pellets fall into the bottle, wash the bottle several times with water.
- Insert a magnetic stir bar into each BOD bottle.
- Insert the sample - pretreated if necessary - into the bottle, measuring its volume with a cylinder.
- If necessary, add ATU and inoculate with a graduated pipette.
- Place the BOD System in the incubator at 20°C
- After 30 minutes of thermostating, screw and tighten the BOD sensors on the bottles and start the test by pressing START
- During analysis, check the software BOD curve with BODSoft™

BMP analysis and anaerobic tests

The preparation of the BMP test is carried out by adding these in the bottles:

- inocula, (par. 12.3) V_i (L)
- nutrient solutions (par. 12.4), V_d (L),
- dilution water, V_H 2 O (L).
- sample, V_c (L or g) (for blanks, this volume is replaced by an equivalent volume of dilution water)

These volumes must be such as to:

- ensure an adequate F / M ratio (FM): the default value suggested for this parameter is, as previously specified, 1 gVS/ gVS and typically varies in the range is [0.25-1];
- operate with an acceptable concentration of volatile solids (inocula + substrate) in the test reactor (C_{max}): an adequate value for this parameter is 10 gVS / L and can vary in the range [1-15];
- measure adequate overpressures, avoiding excessively low increases and greater affects from measurement errors, or ones that are too high which could favour gas leaks or even be incompatible with the available instruments. The maximum overpressure at the end of the test (p_{max}) can be set to 0.4 atm and vary in the typical range [0.2-0.5].

- Set the incubator temperature to the desired value (e.g. 35°C).
- Connect the Wireless DataBox TM and fill in the RESPIROSoft™ software database
- Introduce a magnetic stir bar into each bottle
- Weigh the mass of well mixed inocula in a beaker (helping yourself, in order to be precise in the dosage when approaching the value to be used, with a cylinder or a plastic dropper with the passage hole a little wide so as not to lose representativeness). Note the actually dosed inocula mass and redo the tare.
- Add the nutrient solutions (a volume dosage by cylinder or pipette is fine).
- Add the substrate (except for the "blank" bottles). In the case of sludge, use the same care described in point 4. Note the mass of reagents actually dosed. If you are analysing a powdered substrate such as cellulose, weigh the quantity to be used in a weighing vessel

and pour it into the bottle: to thoroughly clean the cellulose powder left on the vessel, use part of the dilution water (measuring the added volume).

- Check the pH of the mixture and, if lower than 7.3, adjust it, for example with sodium bicarbonate (NaHCO_3). When removing the pH probe from the mix, be careful not to remove too much material from the bottle. In this regard, it is suggested to dose inocula nutrients, substrate and finally only a portion of dilution water, leaving behind about 30 grams which will then be used to rinse the probe. The pH is then measured, if necessary the bicarbonate is added and, subsequently, using the approximately 30 grams of non-dosed water, the probe is rinsed, being sure to remove the settled solids from it, changing them inside the bottle. At the same time, therefore, the dosage of water is completed up to the indicated value (in fact, normally the mains water is slightly alkaline and therefore an incomplete dosage before measuring the pH is good for safety). Write down the test start pH value of each bottle.

- Flush the headspace with nitrogen (or argon) deoxygenated gas: according to the flow for 5 to 10 minutes.

- Introduce about 20 g of KOH into the alkali collector, below the holes. If any KOH pellets fall into the bottle, wash the bottle several times with water.

- Screw the RESPIROMETRIC Sensors onto each bottle and tighten.

- Place the system in the thermostat at the desired temperature, without stirring

- Wait about an hour for the liquid mass present in the bottle to reach the test temperature, and then to develop the typical initial pressure increase, due to the vapor pressure. To remove this additional pressure, it is advisable, after one hour, to vent the test bottles to atmospheric pressure (by briefly loosening the ring nut).

- Start the mixing and data acquisition at the same time.

- The system is ready to start the BMP measurement. Press START and start stirring

The test lasts about 30 days. The test can be considered concluded when the daily production of methane for 3 consecutive days is less than a percentage (1% or 0.5%) of the cumulatively produced methane gas up to that day. It is therefore not a fixed duration: the test time depends on the type of substrate and must be verified during the test itself.

Example: BMP test on cellulose substrate

The inocula and substrate volatile solids content is as follows:

- inocula: 13 gVS/kg
- substrate: microcrystalline cellulose: 950 gVS/kg

The details for setting up the test are shown in the following table:

	BLANK	CELLULOSE
M SUBSTRATE [G]	-	1.2
M INOCULA [G]	175	175
V NUTRIENTS [ML]	Solution A 15 Solution B 15 Solution C 3	Solution A 15 Solution B 15 Solution C 3
V_H2O (ML)	92	90.8
F/M	-	0.5

Table 10

F / M, in terms of volatile solids, is the ratio between sample and inocula (grams of VS of test substance in relation to grams of VS of inocula) (par. 12.13).

With these volumes follow the analytical procedure above.

Determination of the aerobic biodegradability of plastic materials in an aqueous medium

The analysis measures the oxygen demand in a closed respirometer for the determination of the degree of aerobic biodegradability of plastic materials including those containing formulation additives.

The test material is exposed in an aqueous medium under laboratory conditions to an inoculum from activated sludge.

Analysis time: 1 to 6 months

Results: Pressure

Incubation Temperature: 20°C – 35°C

Reference Method: ISO 14851 - Determination of the ultimate aerobic biodegradability of plastic materials in an aqueous medium

Soil quality – Laboratory methods for determination of microbial soil respiration

The analysis with VELD RESPIROMETRIC Sensor provides microbial soil respiration results from the mineralization of organic substances.

A major application of these analyses is related to detection and monitoring processes during decontamination and clean-up processes.

In this process organic substances are oxidized to the end products CO₂ and H₂O, with concurrent uptake of O₂ for aerobic microorganisms.

Analysis time: from 7 to 30 days depending on the substrate

Results: Pressure (mbar)

Incubation Temperature: 20°C – 35°C

Official Reference method:

BS EN ISO 16072 - Soil quality - Laboratory methods for determination of microbial soil respiration

Biodegradation of Monomers

The production of biodegradable plastics has positive repercussions for the environment, but to optimize their management it is necessary to expand the knowledge relating to the conditions in which final degradation can occur with the greatest efficiency.

RESPIROMETRIC Sensors System can be used in order to evaluate the metabolic activity of microbial communities involved in the biodegradation processes of different substrates under varying conditions.

Analysis time: 28 days.

Results: Pressure (mbar)

Incubation Temperature: 20°C

Assessment of the effects of disinfection on BOD determination

In wastewater treatment plants, disinfection is an essential process to produce effluents that can be discharged into the receiving water bodies while minimizing health risk.

The use of disinfectants, however, can affect the Biochemical Oxygen Demand (BOD), both as an interference and for the possible changes in the biodegradability of the organic substance.

Analysis time: 5 days and 28 days.

Results: mg/l and Pressure (mbar)

Incubation Temperature: 20°C

Denitrification activity of the biomass

The VELP RESPIROMETRIC Sensor provides results for the denitrification activity of the biomass.

For the denitrification reaction, the COD/N parameter is defined, which expresses the ratio between the concentration of soluble COD available to heterotrophic bacteria and the concentration of nitrate to be removed.

The COD/N value depends on the type of substrate used and the cell yield coefficient from the biomass.

Analysis time: generally 7 -10 days.

Results: Pressure (mbar)

No incubation required

16. Warranty

The unit is covered by a warranty against manufacturing defects for **25 months** starting from the VELP invoice date.

Any complaints can only be forwarded if the system has been installed according to the specifications of the manual and by qualified and authorized VELP personnel.

By virtue of this guarantee, VELP SCIENTIFICA undertakes to repair the instrument that is defective due to the quality of the material or poor workmanship.

Instruments rendered defective by inexperience and carelessness will not be replaced or repaired.

For further details contact your distributor.

Exclusions:

The warranty is void for defects resulting from:

- inexperience and carelessness on the part of the operator,
- repairs, maintenance or replacements made by personnel or companies not authorized by the manufacturer,
- use of the instrument that does not comply with the instructions / recommendations given in this operating manual,
- use of non-original spare parts.

We would like to receive your valuable suggestions and advice in order to constantly improve these operating instructions

Declaration of conformity

We the manufacturer
address
Velp Scientifica srl
Via Stazione, 16
20865 USMATE (MB)
Italy

under our responsibility declare that the product is manufactured in conformity with the following standards:

EN 61010-1	Safety requirements for electrical equipment for measurement, control, and laboratory use Part 1: General requirements
EN 61326-1	Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 1: General requirements

and satisfies the essential requirements of the following directives:

2006/42/EC	Machinery directive
2014/30/EU	Electromagnetic compatibility directive
2015/863/EU (RoHS III)	Restriction of the use of certain hazardous substances
2012/19/EU (WEEE)	Waste of electric and electronic equipment

Declaration of conformity ^{UK} _{CA}

EN

We the manufacturer
address Velp Scientifica srl
Via Stazione, 16
20865 USMATE (MB)
Italy

under our responsibility declare that the product is manufactured in conformity with the following regulations:

S.I. 2016/1101 Electrical Equipment (Safety) Regulations 2016
S.I. 2016/1091 Electromagnetic Compatibility Regulations 2016

according to the relevant designated standards:

EN 61010-1 Safety requirements for electrical equipment for measurement, control, and laboratory use
Part 1: General requirements
EN 61326-1 Electrical equipment for measurement, control and laboratory use
- EMC requirements -
Part 1: General requirements

and satisfies the essential requirements of regulations:

S.I. 2008/1597 Supply of Machinery (Safety) Regulations 2008
S.I. 2012/3032 Restriction of the Use of Certain Hazardous Substances in
Electrical and Electronic Equipment Regulations 2012
S.I. 2013/3113 Waste Electrical and Electronic Equipment Regulations 2013

Thank you for Thank you for choosing a VELP product!

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VELP Scientifica Srl
Via Stazione, 16
20865 Usmate (MB) - ITALY
Tel. +39 039 62 88 11
Fax. +39 039 62 88 120
www.velp.com



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